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2022 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions

First Practice **Session**

Biochemistry Experiment

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

ÿ All necessary equipment and reagents for the experiment are on the table. Please check them against the list below. If anything is missing...

Please raise your hand to inform the examiners. After the experiment, please clean the used equipment and put it away neatly.

Experimental equipment and reagents:

equipment	Quantity of	experiments and experimental materials	
Micro-dispenser P20	1 piece	Acetone	1 tube
Micro-dispenser P1000	1 piece		
20 µL micropipette tip	1 box of	hexane	1 tube
1000 µL micropipette tip	1 box of	sample A	1 tube
microcentrifuge tubes	8 tubes of	sample B	1 tube
Explosion-proof clip	3 samples	C	1 tube
centrifuge	A thin-layer	chromatography plate	1 sheet
Developing tank (500mL beaker)	1 Groove	ruler	1
Vibrating Mixer	1 pencil		1
Oil-based brush	1 computer		1 unit
Heater (dry bath)	Shared tape station		1 unit

Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. This exam paper (including the cover and the question paper) consists of 6 pages and must be returned in its entirety upon submission.
3. The total time allotted for answering the questions is 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back of the paper, but please indicate the question number.

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I. Experiment Topic: Photosynthetic Pigments

Many organisms on Earth can perform photosynthesis, with the earliest photosynthesis originating from bacteria. In 1988, Nobel... The Bell Prize in Chemistry was awarded to three German scientists, Deisenhofer, Huber, and Michel, in recognition of their contributions to the field. Purple bacteria contribute to the photosynthetic mechanism. The cell membrane of purple bacteria contains bacterial chlorophyll. Bacteriochlorophyll, magnesiobacterial chlorophyll (Bacteriorhodopsin), and some carotenoids. These pigments bind to certain transmembrane proteins, forming antennas that capture light energy or photoreaction processes that convert light energy. The center. However, according to scientists, different organisms use different photosynthetic pigments. Common photosynthetic pigments include chlorophyll a, chlorophyll b, xanthophyll, phycobilins, phycocyanin, and phycoxanthin, among others.

Cells determine which photosynthetic pigments to select, partly due to their evolutionary origin. For example, both cyanobacteria and red algae... It contains chlorophyll a and phycocyanin, therefore the chloroplasts of red algae are believed to have originated from cyanobacteria via endosymbiosis. Evolutionary. Green algae and terrestrial plants both contain chlorophyll a and chlorophyll b. Another category, including diatoms and brown algae, is rich in chlorophyll a, chlorophyll c, and phycoxanthin. Furthermore, photosynthetic pigments are also related to the habitat of organisms. For example, red algae can grow in deeper ocean layers where longer wavelengths of light are less abundant. Therefore, it contains a relatively high amount of phycobiliproteins, proteins that can capture shorter wavelengths of light, and phycoerythrin is one of them. A sort of.

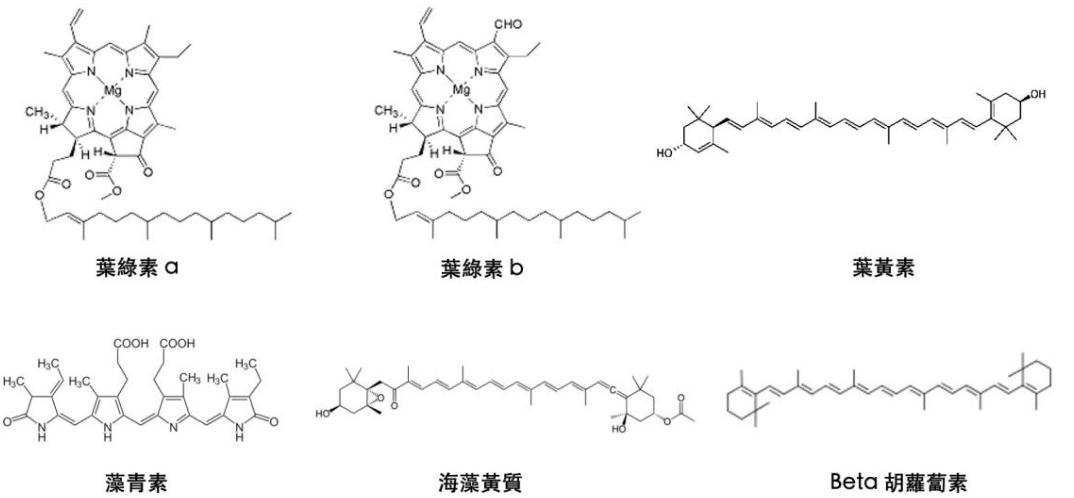

Figure 1: Structure of common photosynthetic pigments

From a chemical structure perspective, photosynthetic pigments all possess many conjugated double bonds, which makes them readily absorb light. Yes. Furthermore, by adding changes to some functional groups, the maximum absorption wavelength of various pigments can be adjusted. For example, leaves... Chlorophyll a and chlorophyll b are very similar in structure, except that chlorophyll b has an additional aldehyde group (Figure). (i) This ultimately results in a difference in the maximum absorption wavelengths of the two (Figure 2). Additionally, photosynthetic pigments also possess structures that help them bind to proteins, such as long hydrocarbon chains providing hydrophobic interactions, or structures that can bind to... Functional groups that form covalent bonds in proteins.

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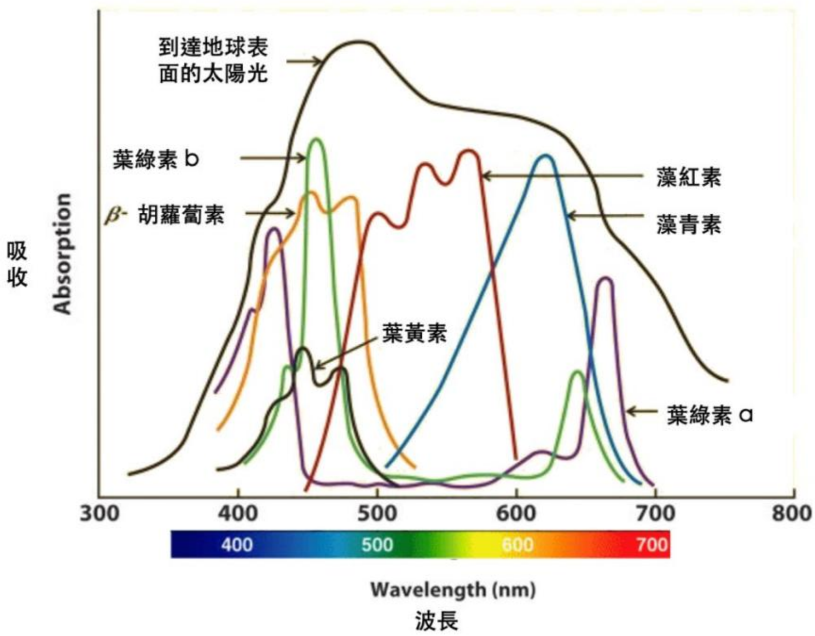

Figure 2: Absorption spectra of common photosynthetic pigments

To analyze the various photosynthetic pigments within cells, the first step is to separate them. Thin-layer chromatography

Chromatography is a commonly used experimental method. You will each receive a thin aluminum sheet uniformly coated with silicone; this silicone layer is called the "static phase." We will then prepare a developing solution as the "mobile phase." The various pigments are separated by the difference in the interaction forces between the static and mobile phases. We can calculate the molecular weight of each component on the thin sheet.

The distance the material moves during chromatography is divided by the distance the front edge of the developing solvent moves to obtain an Rf value (retention value). For example...

Under the same experimental conditions, the same components should have the same Rf value. If we have a certain...

A standard of a pigment can be used to determine which photosynthetic pigment is present in the cell by comparing Rf values.

In this experiment, you will obtain tissue samples from three types of algae: red algae, green algae, and diatoms.

Please extract the photosynthetic pigments from algae according to the instructions, analyze the photosynthetic pigments contained in the samples using thin-layer chromatography, determine which type of algae each sample belongs to, and answer the experimental questions.

II. Experimental Procedure: (10 **points)**

2-1 Algal Sample Processing:

1. First observe the appearance of the three algal samples A, B, and C, and record your observations on page 5.
2. Add 1ml of acetone to each ingredient, close the lid tightly and attach the explosion-proof clip, and shake in a shaker for 2 minutes.
3. Place the sample in an 80-degree heater and heat for 5 minutes.
4. After heating, insert the sample into an ice bucket to cool for 1 minute, then shake it for 1 minute using a shaker.
5. Remove the explosion-proof clips, place the samples symmetrically into a benchtop centrifuge, and centrifuge for 5 minutes at room temperature.

   Bell (Little Turtle Centrifuge).
6. After centrifugation, use a micro-dispenser P1000 to aspirate 700 µL of supernatant.
7. Observe the colors of the supernatant and precipitate and record them on page 5.

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2-2 Thin-layer chromatography pigment analysis:

1. Take a thin-layer chromatography slide, white side up, and at one end of the short side, 1 cm from the edge, use...

   Draw a line with a pencil as the starting line. Then, on the starting line, draw three short lines vertically, each approximately [length missing].

   0.5 cm apart, spaced 1.5 cm apart, and labeled with the exam number (as shown in Figure 3).

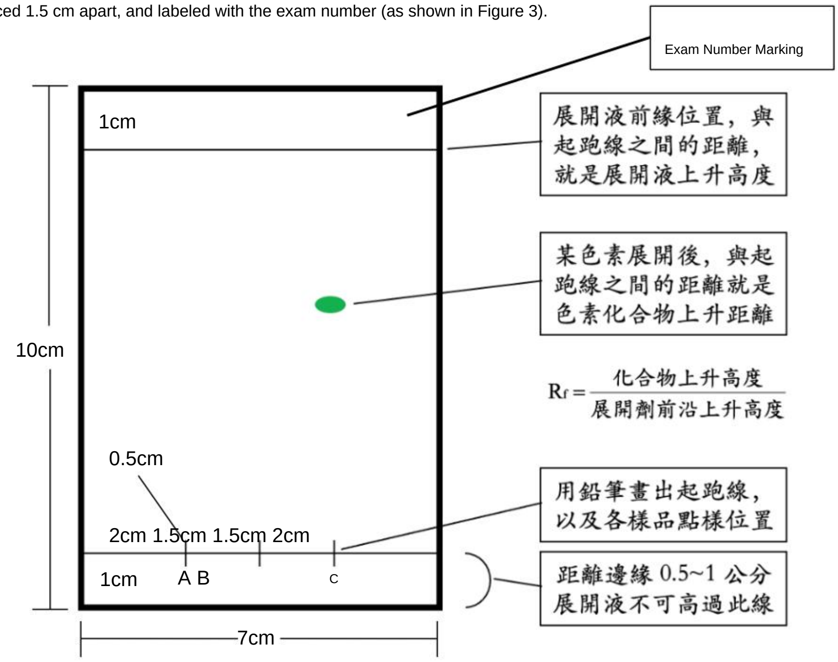

Figure 3: Schematic diagram of thin-layer chromatography

2. Take 100 µL of the supernatant sample from step 2-1-6 and dispense it into a P1000 micro-dispenser.

   Prepare new microcentrifuge tubes (labeled A, B, C) for later use.

3. How to: At the dot on the thin laminizer, 4 ÿL x 20 were taken for each group with a microdispenser P20

   Apply the supernatant (80 µL total) to the sample.

   Tips

   i.The supernatant will first diffuse after the point goes up, and then the acetone can be repeated after the volatilization

   ii. To ensure better resolution in thin-layer chromatography, the diameter of the sample spot should not exceed 0.6 mm.

   centimeters

   iii.Draw 3 dot positions at the starting line, marking samples A, B, and C with pencil respectively.

4. Prepare the developing solution: Mix 8 ml of acetone and 12 ml of hexane and pour the mixture into the developing tank.

5. Place the spotted thin-layer chromatography slide into the developing tank, ensuring the developing solution level does not exceed the starting line.

   Cover the expansion slot with plastic wrap.

6. After approximately 20-25 minutes, when the leading edge of the developing solution is about 1 cm above the surface, the process is complete. Use a pencil.

   Mark the leading edge position, remove the thin-layer chromatography slide, and observe the results after the solvent has evaporated and the slide has dried.

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III. Experimental Results (50 points)

1. Sample observation (9 points, 1 point per square)

	Sample A	Sample B	Sample C
Original sample color			
Supernatant color			
precipitate color			

2. How to result: What are the diameters of each of your sets of samples how, and please briefly describe the process of how you. (12 points)

3. Please attach the thin-layer chromatography plate to the bottom with tape and calculate the Rf value of all green pigments in each group of samples.

(9 points, 3 points per square)

	Sample A	Sample B	Sample C
Write down the contents of the sample from top to bottom. Some green pigments have Rf values			

Please attach the thin-layer chromatography plate to the bottom with tape: Experimental Results (20 points)

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IV. Questions and Answers (50 points)

1. Which green pigment is present in all samples (please write its Rf value)? What do you think it is?

Photosynthetic pigments? (10 points)

2. Which pigments have the worst absorption capacity for orange-yellow light? (Please write down which sample and what the Rf value of the pigment is.)

(5 points) What type of photosynthetic pigment do you think it is? (10 **points)**

3. Why, after acetone extraction, does a large amount of the red pigment still remain in the precipitate? Please explain.

Provide possible explanations. (10 points)

4. Based on the experimental results, what type of algae are samples A, B, and C? (5 **points)** Your judgment is based on...

What is "ÿ"? (10 points)